Kavya Karunakaran

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WORK EXPERIENCE

Software Engineer I Jul 2025 - Present
Oracle Corporation *Redwood Shores, CA*

- Shipped Java enhancements for Oracle's NetSuite ERP, including order allocation and inventory management integrations, navigation, and bulk selection, now used by approximately **4,400 customers** (10% of NetSuite's customer base).
- Investigated and resolved production issues in inventory allocation, transaction processing, and revenue recognition using Java, SQL, and Timber logs, including root-causing a **high-severity customer escalation** under a 24-48 hour deadline.
- Authored and deployed **SQL data migrations** and corrective scripts for customer data issues, including one that restored **2M+ records**. Collaborated with engineering, QA, DevOps, product, and support teams.
- Implemented security fixes for RESTful API request vulnerabilities and SQL injection risks.
- Wrote **30+** Playwright and JUnit tests, including revenue commitment and reversal UI tests, achieving **100% coverage** for each feature tested. Tests ran in TeamCity and were monitored in Timber.
- Used AI-assisted development tools for debugging, root-cause analysis, and code generation, validating output through code review and regression testing.

Software Engineering Intern Jul 2024 - Sep 2024
Truveta *Bellevue, WA*

- Built and automated a **distributed data-validation pipeline** using Apache Spark, Python, and Azure, processing and validating **10M+ records**.
- Ingested Parquet files from Azure Storage into DataFrames, hashed and decrypted IDs using Azure Key Vault secrets, transformed data through filtering and joins, and wrote results back to Azure Storage.
- Used Azure Managed Identities and Federated Credentials to access Key Vaults and storage accounts, with Azure DevOps Pipelines, Netflix Conductor, and Datadog supporting development and testing workflows.

Software Engineering Intern Jun 2023 - Jun 2024
CalMatters, Digital Democracy Project *San Luis Obispo, CA*

- Wrote a script that ran daily to cleanse a MySQL organizations table of **approximately 11,000 records**, removing bad data and merging duplicates; used Pandas to extract and analyze data from Google Sheets.
- Scraped and parsed Word docs with Python, then used fuzzy string matching to extract and validate organization names.

EDUCATION

California Polytechnic State University, San Luis Obispo Jun 2025
Bachelor of Science in Computer Science GPA: 3.5/4.0

Coursework: Database Modeling Design and Implementation, Distributed Computing, Knowledge Discovery from Data, Deep Learning, Applied Linear Models, Introduction to Computer Security
Leadership: President, A Cappella; Campus Officer, Women Involved in Software and Hardware (WISH)

PROJECTS

Predicting Tipsheet Newsworthiness Scores with Machine Learning (CalMatters Research Project) Sep 2024 - Jun 2025

- Built a natural language processing pipeline with NLTK and Scikit-learn to predict newsworthiness scores of legislative tipsheet transcripts, applying stemming, stop-word removal, and class balancing to prepare training data.
- Trained and evaluated Naive Bayes, random forest, support vector machine, and boosting classifiers on **approximately 6,300 transcripts** using cross-validation and confusion matrices, achieving **82.08% accuracy**.

Knowledge Discovery from Data Sep 2023 - Dec 2023

- Implemented data mining and machine learning algorithms in Python: PageRank, collaborative filtering, hierarchical clustering, random forest classification, and cross-validation; evaluated algorithm performance.
- Performed dataset analysis using association rule mining and agglomerative and KMeans clustering.

SKILLS

Languages: Java, Python, SQL, Go, R, C, JavaScript, HTML, CSS, RISC-V
Backend & Testing: Backend Development, RESTful APIs, Distributed Systems, JUnit, Playwright
Data & Databases: MySQL, Apache Spark, Pandas, NumPy, Data Pipelines, Data Migration, Data Validation, Data Mining
Cloud & DevOps: Azure (Blob Storage, Key Vault, DevOps), Netflix Conductor, Datadog, TeamCity, Timber, Git, Linux, CI/CD
Fundamentals & Process: Object-Oriented Programming, Data Structures & Algorithms, Agile, Scrum, Jira
Machine Learning: PyTorch, Scikit-learn, NLTK, Machine Learning, Deep Learning, Natural Language Processing